Adversarial Logit Separation

by Zixi Chen, Jinli Xiao, Yifei Zhu, Jiawei Zhang

NYU

Content

- Methodologies
- Theoretical Foundations
- Experimental Results

Methodologies

Overview

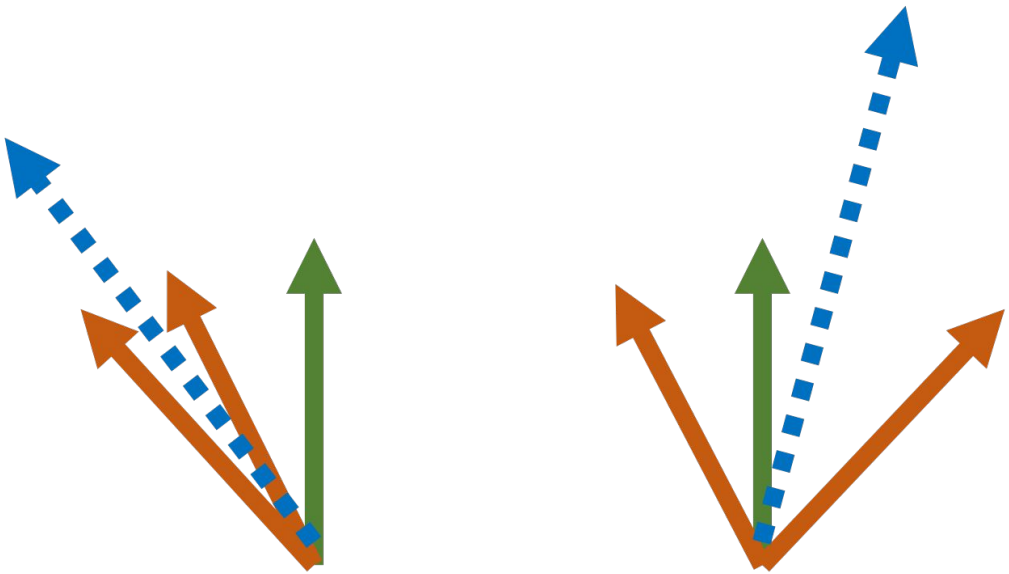
- A loss component base on the notion of *logit separation*.
- Sequentially train several parameter-diverse models.

$$f_1, f_2, \cdots, f_I \qquad h = \sum_{i=1}^{I} \alpha_i f_i.$$

- Prevent overfitting by using simulated annealing scheduler.

Logit Separation - Intuition

- Models with larger distances between logits indicates diversity
- Ensembling diverse models yields better robustness.
- The general idea for logit separation is to force diversity.



1. The green arrow is the true label.
2. The orange arrow is the adversarial perturbed vectors.
3. The blue arrow is the ensemble vector

Logit Separation - Loss Component

- **The idea is to penalize the cosine distance of non-true logits**

$$m = \mathbf{1} - y = [1, \cdots, 0, \cdots, 1]^T$$

$$L_{\text{sim}}(f_r(\tilde{x}), f_o(\tilde{x}), y) = \lambda \cdot \frac{(f_r(\tilde{x}) * m)^T (f_o(\tilde{x}) * m)}{||f_r(\tilde{x}) * m|| \cdot ||f_o(\tilde{x}) * m||}$$

where the 0 is the y-th (label) index in m, the mask

where f_r is the reference model and f_o is the other models.

Logit Separation - Total Loss Function

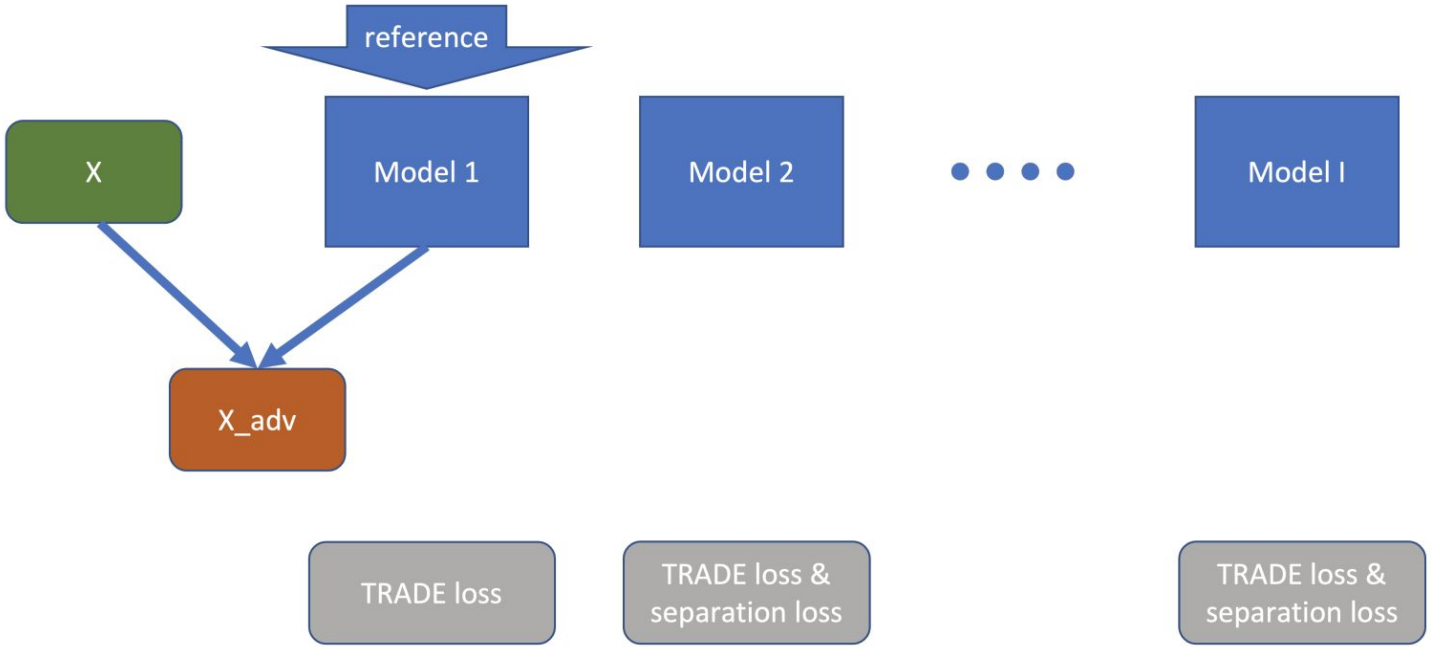
$$L = \mathop{\mathbb{E}}_{(x,y)\sim D}[\phi(f(x)y) + \lambda_1 \cdot \phi(f(\tilde{x})f(x)) + \lambda_2 \cdot L_{\text{sim}}(f_r(\tilde{x}), f_o(\tilde{x}), y)]$$

where ϕ is a surrogate 0-1 loss function.

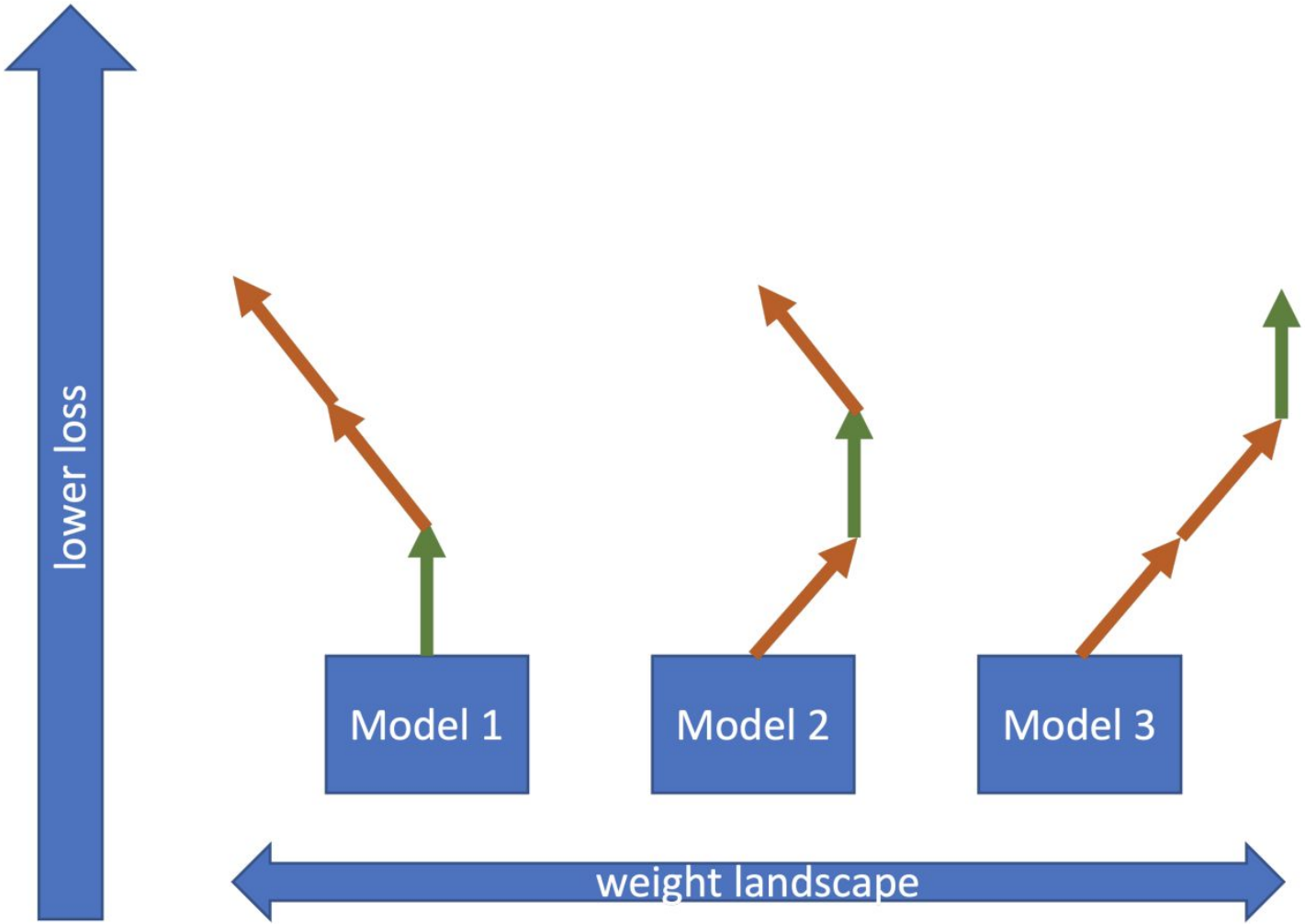
Here we use TRADES (Zhang et al., 2019) as our base model

Ensemble Iterative Method, Part I, Graphic illustration

- **Reference model is selected in turn from each model.**
- **Reference generates adversarial examples to all others.**

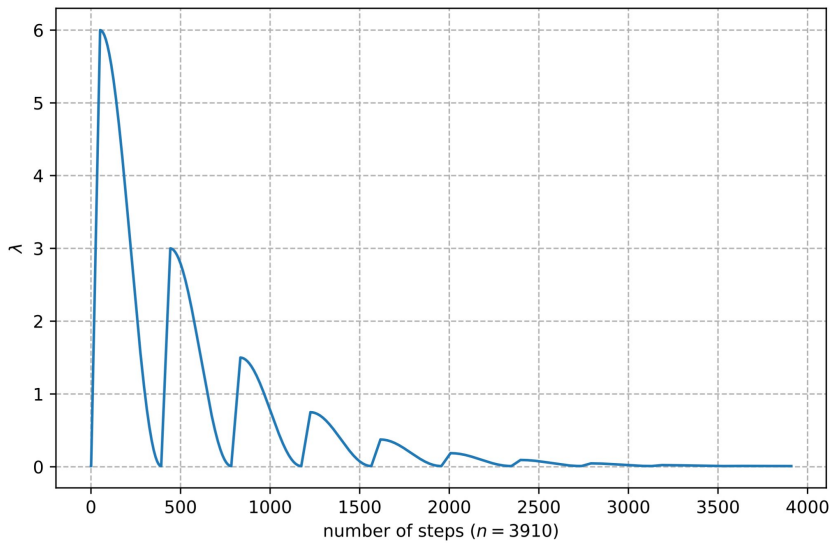


Ensemble Iterative Method, Part II, Graphic Illustration



Cosine Annealing Scheduler

$$L = \mathop{\mathbb{E}}_{(x,y)\sim D}[\phi(f(x)y) + \lambda_1 \cdot \phi(f(\tilde{x})f(x)) + \lambda_2 \cdot L_{\text{sim}}(f_r(\tilde{x}), f_o(\tilde{x}), y)]$$



Algorithm Pseudocode

Algorithm 1 Logits Separation Joint Training

```
for i ← 1 to I do                          ▷ I is the number of models
    init f_i
end for
shr ← CosineAnnealingScheduler()
repeat
    for (x, y) ← 1 to n do                 ▷ n is number of batches
        for r ← 1 to I do                  ▷ f_r s be reference model in turn
            λ ← shr.step()
            x̃ ← PGD(f_r, x)
            for o ← 1 to I do
                L ← TRADES_loss(x, x̃, y)
                m ← 1 − y
                L ← L + 1_{r≠o} λ · (f_r(x̃)·m)^T (f_o(x̃)·m) / (||f_r(x̃)*m|| · ||f_o(x̃)*m||)
                Conduct gradient descent on loss L for f_o
            end for
        end for
    end for
until λ converges
return ensemble of models
```

Comparison

$$L_{f_o} = \mathop{\mathbb{E}}_{(x,y)\sim D}[\phi(f_o(x), y) + \lambda_1 \cdot \phi(f_o(\tilde{x}) f_o(x)) + \lambda_2 \cdot L_{\text{sim}}(f_r(\tilde{x}), f_o(\tilde{x}), y)]$$

Logit Separation

TRADES

$$L = \phi(f(X), Y) + \lambda\phi(f(X) f(\tilde{X})$$

(Zhang et al., 2019)

ADP

$$L = \sum \phi(f_k, X, y) + \alpha H(\mathcal{F}) + \beta \log\left(\sum_{i \neq j} (f_i(X))^T f_j(X)\right)$$

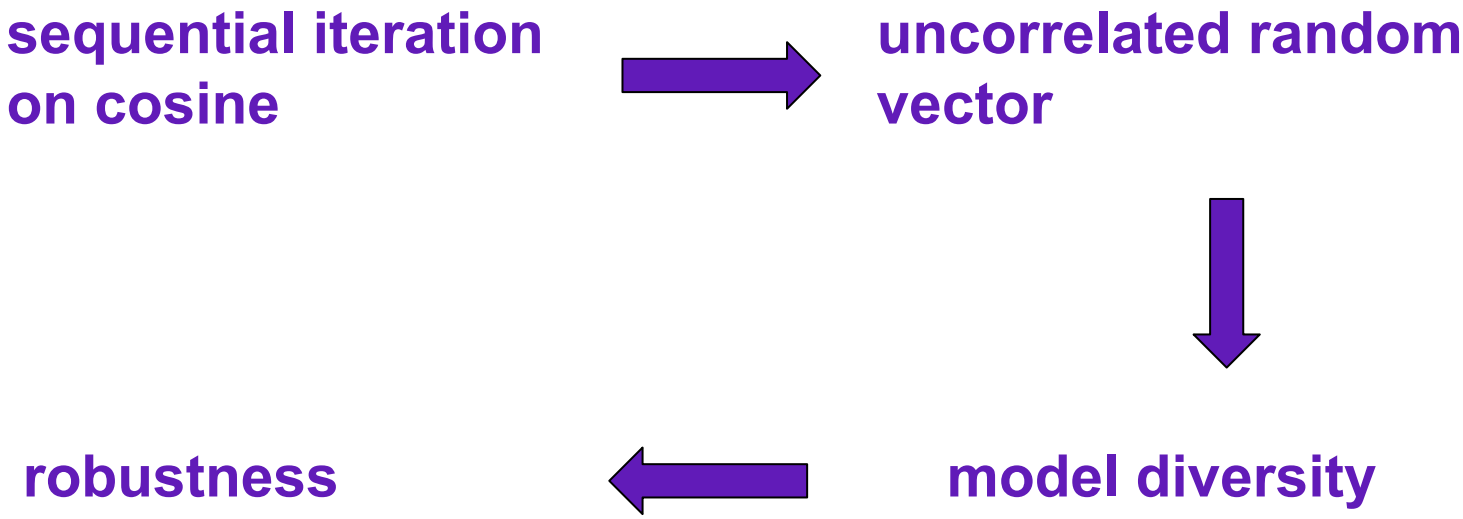
(Pang et al., 2019)

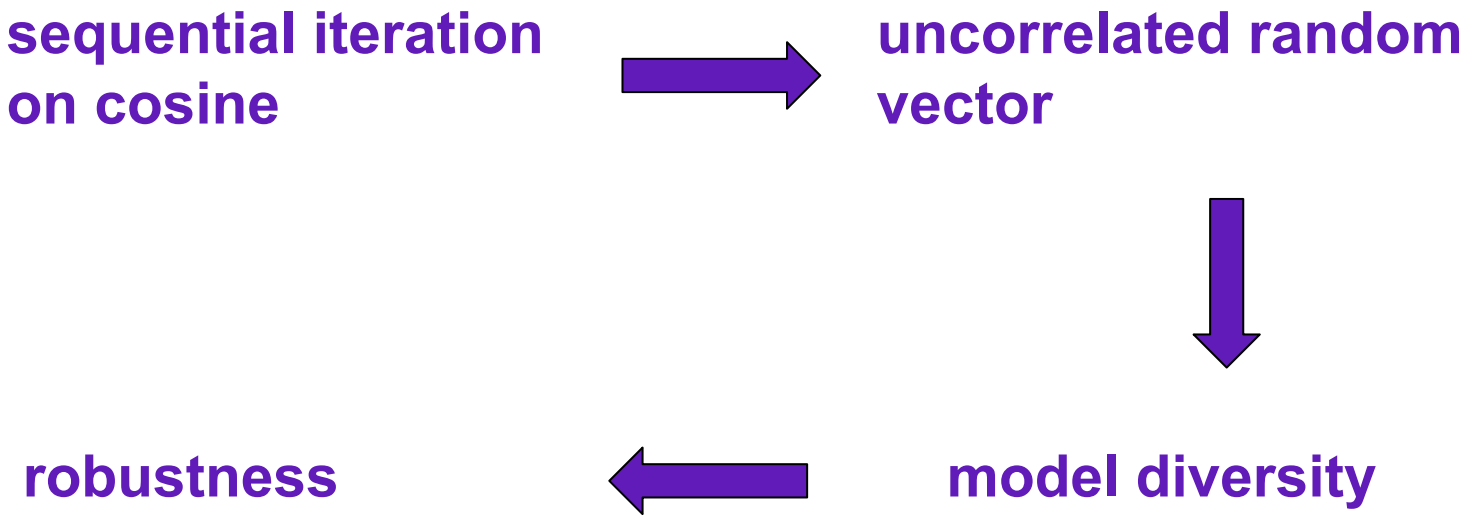
ALP

$$L = t\phi(f(X), Y) + (1-t)\phi(f(\tilde{X}, Y) + \lambda\phi'(f(\tilde{X}), f(X))$$

(Kannan et al., 2018)

Theoretical Foundations


sequential iteration on cosine
uncorrelated random vector
model diversity
robustness


sequential iteration on cosine
uncorrelated random vector
model diversity
robustness

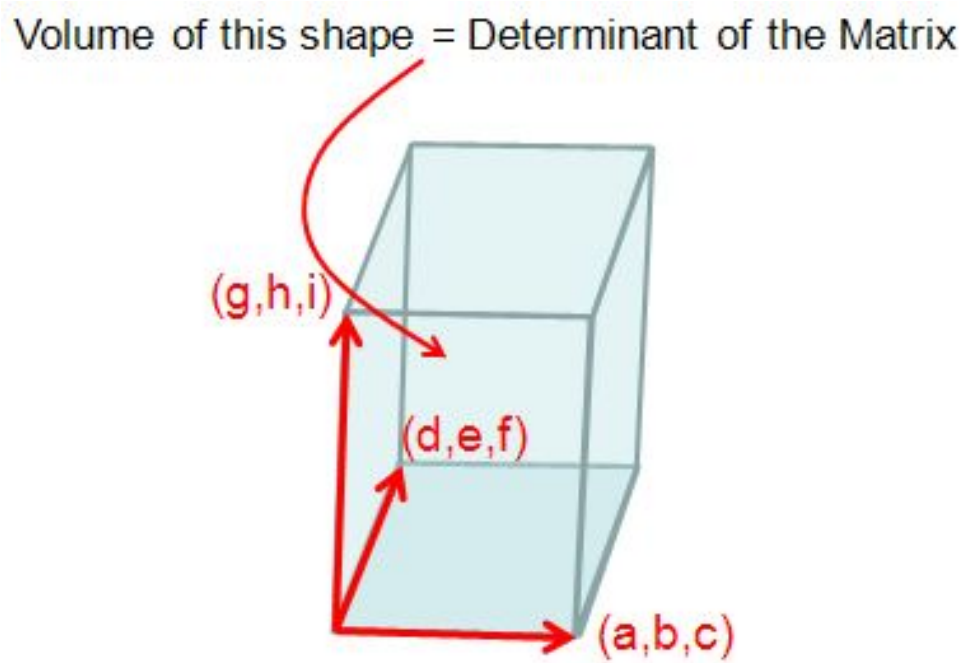
Logit separation definition

$$m = \mathbf{1} - y = [1, \cdots, 0, \cdots, 1]^T$$

where the 0 is the yth (label) index

$$L_{\text{sim}}(f_r(\tilde{x}), f_o(\tilde{x}), y) = \lambda \cdot \frac{(f_r(\tilde{x}) * m)^T (f_o(\tilde{x}) * m)}{||f_r(\tilde{x}) * m|| * ||f_o(\tilde{x}) * m||}$$

Pang et al., 2018: $$\det(\tilde{M}_{\setminus y}^{\top}\tilde{M}_{\setminus y}) = \mathrm{Vol}^2(\{\tilde{F}_{\setminus y}^{k}\}_{k\in[K]}),$$



https://www.sharetechnote.com/html/Handbook_EngMath_Matrix_Determinent.html

det v.s. sum of cosines

Elegant geometric interpret & efficient computation cost

simultaneous v.s. sequential

Following the same proof paradigm as Pang:

We define

$$a_{i,j} = \{(f_r(\tilde{x}))_i = (f_o(\tilde{x}))_j\}$$ **(hyperplane)**

$$A_{i,j} = \{(f_r(\tilde{x}))_i = (f_o(\tilde{x}))_j + c, c \in \mathbb{R}\}$$ **(affine plane)**

$$a^+_{i,j} = \{(f_r(\tilde{x}))_i \geq (f_o(\tilde{x}))_j\}$$ **(half space)**

Decision region:

$$A_{\hat{y}} = \bigcap_{i \neq \hat{y}} a^+_{\hat{y},i}$$

And our targeted solution space $S_{\hat{y}}$ is:

$$S_{\hat{y}} = (\bigcap_{i,j \neq \hat{y}} a_{i,j}) \cap A_{\hat{y}}$$

Result:

In the decision space $A_{\hat{y}}$ of class $\hat{y}$,

$\forall i, j \neq \hat{y}, z_{\hat{y}} \in A_{\hat{y}}$, we have a unique

$a^0_{i,j} \in A_{i,j}$ s.t. $z_{\hat{y}} \in \bigcap_{i,j \neq \hat{y}} a^0_{i,j} = Q_0$.

And the solution set to the problem is:

$$\operatorname{argmin}_{z_{\hat{y}}}(\max_{z \in Q_0 \cap \bar{A}_{\hat{y}}} (f(z))_{\hat{y}}) = S_{\hat{y}}.$$

Experimental Results

Ensemble Result

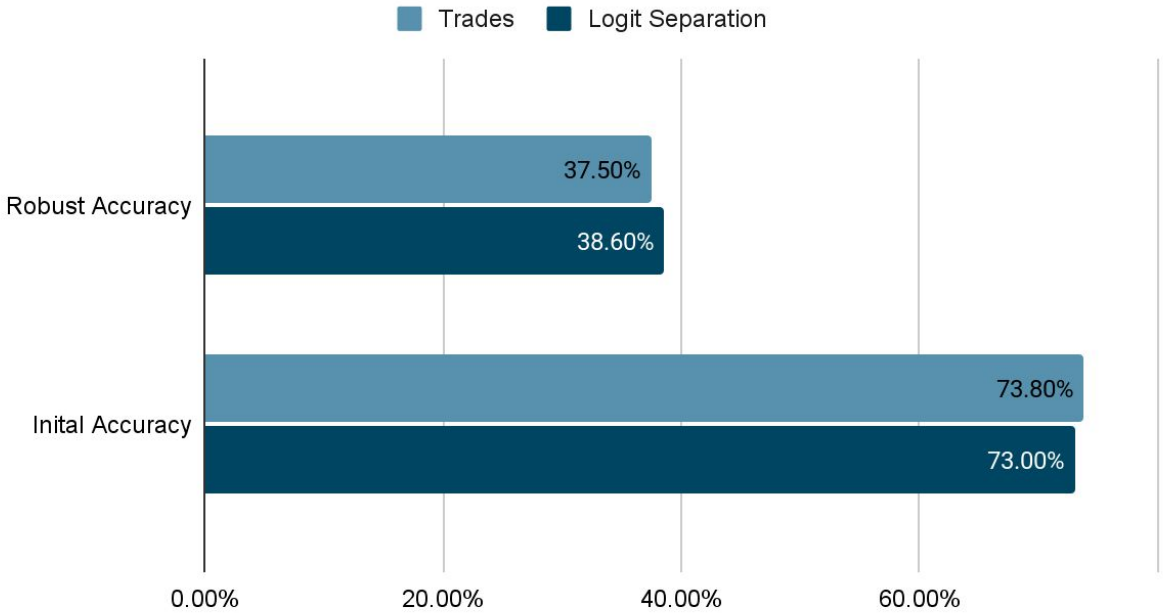
	clean data	PGD adversarial data
Baseline	0.43087897	0.4294527
Ours	**0.4010905**	**0.3988733**

Cosine distance of logits.

		model 1	model 2	model 3
Baseline	initial	73.50%	**74.30%**	73.90%
	adversarial	36.90%	35.30%	36.40%
Ours	initial	72.80%	72.90%	73.00%
	adversarial	37.00%	**37.50%**	37.20%

Accuracy of individual models on clean and adversarial data (autoattack)

Comparison



	Initial accuracy	autoattack accuracy
Baseline	**73.80%**	37.50%
Ours	73.00%	**38.6%**

Accuracy of the ensemble on clean and adversarial data (autoattack)

Conclusion Part

Conclusion

- Use logit separation to improve adversarial robustness
- Can substitute other base models as long as it has a loss function related to robustness.
- We showed the improved performance of TRADES and gave theoretical or intuitive analysis on many assumptions we made.
- More experiments and ablation studies are required.

Q&A

References

- Athalye, A., Carlini, N., and Wagner, D. Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples. In International conference on machine learning, pp. 274–283. PMLR, 2018.
- Carlini, N. and Wagner, D. Towards evaluating the robustness of neural networks. In 2017 ieee symposium on security and privacy (sp), pp. 39–57. IEEE, 2017. Covington, P., Adams, J., and Sargin, E. Deep neural networks for youtube recommendations. In Proceedings of the 10th ACM conference on recommender systems, pp. 191–198, 2016.
- Croce, F. and Hein, M. Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks. In International conference on machine learning, pp. 2206–2216. PMLR, 2020. Croce, F., Andriushchenko, M., Sehwag, V., Debenedetti, E., Flammarion, N., Chiang, M., Mittal, P., and Hein, M. Robustbench: a standardized adversarial robustness benchmark. In Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track, 2021. URL https://openreview.net/forum? id=SSKZPJCt7B.

References

- Engstrom, L., Ilyas, A., and Athalye, A. Evaluating and understanding the robustness of adversarial logit pairing. arXiv preprint arXiv:1807.10272, 2018.
- Eykholt, K., Evtimov, I., Fernandes, E., Li, B., Rahmati, A., Xiao, C., Prakash, A., Kohno, T., and Song, D. Robust physical-world attacks on deep learning visual classification. In Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 1625–1634, 2018.
- He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 770–778, 2016.
- Kannan, H., Kurakin, A., and Goodfellow, I. Adversarial logit pairing. arXiv preprint arXiv:1803.06373, 2018.
- Kurakin, A., Goodfellow, I., Bengio, S., et al. Adversarial examples in the physical world, 2016.

References

- Morgan Kaufmann. Liu, Y. and Yao, X. Ensemble learning via negative correlation. Neural networks, 12(10):1399–1404, 1999.
- Madry, A., Makelov, A., Schmidt, L., Tsipras, D., and Vladu, A. Towards deep learning models resistant to adversarial attacks, 2017. URL https://arxiv.org/ abs/1706.06083.
- Pang, T., Du, C., Dong, Y., and Zhu, J. Towards robust detection of adversarial examples. Advances in Neural Information Processing Systems, 31, 2018.
- Pang, T., Xu, K., Du, C., Chen, N., and Zhu, J. Improving adversarial robustness via promoting ensemble diversity. In International Conference on Machine Learning, pp. 4970–4979. PMLR, 2019.

References

- Szegedy, C., Zaremba, W., Sutskever, I., Bruna, J., Erhan, D., Goodfellow, I., and Fergus, R. Intriguing properties of neural networks. arXiv preprint arXiv:1312.6199, 2013.
- Tramer, F., Kurakin, A., Papernot, N., Goodfellow, I., ` Boneh, D., and McDaniel, P. Ensemble adversarial training: Attacks and defenses. arXiv preprint arXiv:1705.07204, 2017.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. Advances in neural information processing systems, 30, 2017.
- Zhang, H., Yu, Y., Jiao, J., Xing, E., El Ghaoui, L., and Jordan, M. Theoretically principled trade-off between robustness and accuracy. In International conference on machine learning, pp. 7472–7482. PMLR, 2019.